Note on Week 3: Practice session was not held; Weekly Record 3 was officially waived under course announcement.

APPENDIX

Student: Manat Amir
Group: DPA-2602
Course: Business Administration (Assignment 1)

1. Artifact Week 1: Environment and AI-opportunity map

Environmental & AI-Opportunity Matrix for AB Auto:
Sector: Political / Legal & Fiscal
Macro & Local Operational Reality: Strict revenue turnover ceilings separating Individual Entrepreneurship (IP) simplified regimes from corporate VAT/LLP (TOO) status.

Direct Impact on AB Auto: Business splits operations across four individual entrepreneurships to remain within simplified tax thresholds legally.

Potential AI / Digital Opportunity: Automated accounting and revenue-tracking scripts to balance turnover limits across the 4 IP entities in real time.

Sector: Technological & Financial
Macro & Local Operational Reality: Universal adoption of Kaspi ecosystem (Kaspi QR, Kaspi Pay, Kaspi Shop); payment commission differentials between cash and QR.

Direct Impact on AB Auto: Dual-pricing mechanisms (cash discounts vs. QR markup to cover banking fees); manual paper ledger recording of daily sales.

Potential AI / Digital Opportunity: Computer vision or optical character recognition tools to digitize handwritten sales logs into structured inventory databases.

Sector: Social & Cultural
Macro & Local Operational Reality: High value placed on informal rapport, egalitarian interaction ("brotherhood" ethos), and personal trust in Kazakhstan retail.

Direct Impact on AB Auto: Counter staff act as trusted advisors rather than transactional clerks; formal European-style sales protocols are culturally rejected.

Potential AI / Digital Opportunity: Conversational AI (WhatsApp voice-note transcription) that maintains an informal, localized tone for incoming quote requests.

Sector: Competitive
Macro & Local Operational Reality: Direct competition from bazaar stalls, independent retail shops, and formal platforms like Kaspi Shop and Interkom.

Direct Impact on AB Auto: Necessity of fast fulfillment for commercial fleets where vehicle downtime represents substantial daily revenue loss.

Potential AI / Digital Opportunity: Predictive stock replenishment algorithms based on fleet breakdown patterns (seasonal heating/cooling components)

2. Artifact Week 2: Stakeholder and ethics map
Comprehensive Stakeholder Matrix for AB Auto:

Internal Stakeholders: Owner-Operator (Founder) who directs strategic purchasing; Frontline Sales & Inventory Staff who execute product intake and maintain personal counter relationships.

External Primary Stakeholders: Partnering Auto Repair Stations (STOs) utilizing high-trust informal trade credit recorded in ledger notebooks; Independent Commercial LCV Drivers; Corporate Fleet Clients (e.g., Stolichny Khleb LLP) requiring strict B2B invoice documentation; Upstream Multi-Tier Suppliers (Interkom, Satis Auto, Planeta Auto).

External Secondary Stakeholders: State Revenue Committee (KGD) enforcing fiscal transparency; Public Safety & Road Users relying on component durability.

Focal Tension Analysis: AB Auto operates within an acute tension between informal, localized trust mechanisms and rigid formal institutional compliance. Cross-border freight transit variances force AB Auto to balance certified Russian OEM components (KraftTech) against imported Chinese equivalents (LYNX) to maintain immediate customer cash flow versus ensuring maximum long-term roadworthiness.

3. AI-Use Declaration (Level 2: Supported)
As required by the course Academic Integrity policy for AI-supported tasks.

AI System Used: Google Gemini (Google AI Pro).
What it was used for: Brainstorming, structural feedback, language support, and mapping raw operational data to the OpenStax academic taxonomy.

What was verified independently: All factual claims, quantitative data, and operational mechanics regarding AB Auto, the 4-IP taxation structure, STO ledger trade credit, supply chain delays (1-3 days), and specific component brands (LYNX, KraftTech). Verified against personal operational experience and local market realities.

What was changed as a result: Rejected generic AI recommendations in favor of localized, Kazakhstan-specific strategic actions (e.g., hybrid STO record-keeping instead of heavy corporate ERP systems). Authored the final transfer judgments

ASTANA IT UNIVERSITY

School of Digital Public Administration

WEEKLY INTEGRATED LEARNING RECORD 4

Student Name: Manat Amir
Group: DPA-2602
Course: Business Administration (BA 1204)
Academic Year: 2026–2027
Focal Org (Company B): AB Auto (Astana, Kazakhstan)
Comparator Org (Company A): Interkom-Auto (Russia / CIS)
Topic: Management & Leadership (Ch. 6–7)
Attestation Period: 1st Attestation

FIELD 1 · CONCEPT (CART Lens: Concept)

Management is the structured process of coordinating physical, human, financial, and informational resources through planning, organizing, leading, and controlling to accomplish organizational objectives effectively and efficiently [1]. **Planning** defines future organizational trajectories, resource allocation parameters, and risk horizons. **Organizing** structures authority lines, departmentation, and physical workflow arrangements. **Leading** guides, inspires, and directs operational teams toward shared operational targets. **Controlling** measures empirical performance against baseline standards and triggers corrective interventions [1].

Concept Check — "Can an AI system be accountable for an operational failure?": No. In managerial and legal theory, accountability requires legal agency, moral responsibility, and contractual liability—attributes exclusive to human managers [3]. While artificial intelligence executes automated tasks, detects inventory patterns, and processes unstructured datasets, accountability remains strictly non-delegable. When an algorithmic pricing rule fails or inventory runs out, liability rests entirely with the designated human owner who approved the system guardrails, oversight thresholds, and fallback contingencies [3], [4].

FIELD 2 · APPLICATION (Human–AI Task Allocation Matrix)

The following matrix maps eight critical managerial tasks for AB Auto across the four management functions, categorized via the five operational tests (Accountability, Cost of Error, Empathy/Trust, Volume/Rules, Verification), with explicit human roles and Company A transfer judgments.

#	Function	Task (Verb + Object)	Category	Justification (Formula)	Accountable Role	Transfer Verdict
1	**Planning**	Establish retail price floors on Kaspi marketplace	**Human + AI**	Human + AI because of accountability — owner enforces strict 18% gross margin limits	General Manager / Owner (Top)	Adapt: Adapt Interkom dynamic repricing with owner margin limits.

				across 4 IPs [2].		
2	**Planning**	Forecast seasonal demand for GAZelle suspension kits	**AI Executes + Human Review**	AI Executes + Review because of volume and rules — managing >3,000 SKUs requires ingestion of seasonal road-thaw trends [1], [2].	Purchasing Lead (Middle)	Adapt: Retrain Interkom forecasting logic for Astana fleet cycles.
3	**Organizing**	Ingest vendor invoices via OCR and generate catalog drafts	**AI Executes + Human Review**	AI Executes + Review because of cost of error — SKU mismatch corrupts warehouse records [1], [2].	Senior Storekeeper [CRITICAL REVIEW] (Supervisory)	Adopt: Directly adopt Interkom paper invoice OCR scanning pipeline.
4	**Organizing**	Maintain visual bin layout & heavy parts at Koktal hub	**Human Only**	Human Only because of verification — heavy leaf springs require tactile physical staging [1].	Warehouse Specialist (Supervisory)	Reject: Interkom automated robotic sorting is cost-prohibitive for B.
5	**Leading**	Negotiate informal credit lines with partner STO mechanics	**Human Only**	Human Only because of empathy and trust — competitive moat relies on personal trust ledgers [1], [4].	General Manager / Owner (Top)	Reject: Interkom automated banking checks destroy relationship trust.
6	**Leading**	Coach counter staff on digital cross-referencing	**Human + AI**	Human + AI because of accountability — store managers coach clerks using AI search tools [3], [4].	Store Manager (Middle)	Adapt: Adapt Interkom corporate training into WhatsApp micro-modules.
7	**Controlling**	Delist zero-inventory SKUs on Kaspi automatically	**Automated**	Automated because of volume and rules — rules-based database trigger hides sold-out items instantly [2].	E-Commerce / IT Lead (Middle)	Adopt: Adopt Interkom real-time API inventory cutoff triggers.

8	**Controlling**	Evaluate customer warranty defect claims	**Human + AI**	Human + AI because of cost of error — AI checks warranty terms, but manager inspects metal fractures [1], [2].	Store Manager (Middle)	Adapt: Adapt Interkom warranty portal to a mobile photo desk.

FIELD 3 · VERIFICATION AND AI USE (CART Lens: Reflection)

• **Quantitative Claim Verified:** AB Auto actively maintains >3,000 distinct commercial SKUs across its Koktal 15B warehouse and Abylai Khan 62 branch [1], [2].
• **Verification Method:** Cross-referenced against warehouse bin count logs and 1C trade ledger balances.
• **Discrepancy & Correction:** Initial generative suggestions recommended fully automating customer credit scoring. This was rejected: AB Auto operates across 4 Individual Proprietorships (IPs) relying on localized trust and debt notebooks with fleet mechanics. Enforcing rigid automated credit scoring would alienate key clients and disrupt working capital.
• **AI-Use Disclosure:** Google Gemini (Level 2 Supported) was utilized for brainstorming task allocation categories and structuring the five operational tests. All organizational data, role assignments, and final judgments were authored and verified by the student.

FIELD 4 · TRANSFER JUDGMENT (CART Lens: Transfer)

1. Vendor Invoice OCR Pipeline (ADOPT): AB Auto should directly adopt Interkom-Auto's automated OCR intake of vendor manifests to eliminate clerical data entry errors [2].
2. Automated Conveyor Sorting (REJECT): Interkom automated sorting lines must be rejected due to physical warehouse floor space constraints and the high unit weight of commercial leaf springs and gearboxes [1].
3. Algorithmic Credit Scoring (REJECT): Wholesale adoption of Interkom bank-integrated credit scoring is rejected to protect AB Auto core competitive moat—informal counter trust with Astana mechanics [4].
4. WhatsApp Micro-Training Modules (ADAPT): Interkom lengthy desktop LMS manuals should be adapted into short, mobile-friendly WhatsApp lookup checklists for frontline counter staff [3].

FIELD 5 · EVIDENCE (IEEE Reference List)

[1] L. J. Gitman, C. McDaniel, A. Shah, et al., Introduction to Business, 2nd ed. Houston, TX: OpenStax, Rice University, 2026, ch. 6, 7.
[2] G. Yeshenkulova, N. Ibadildin, and M. Tleubayeva, Business Administration. Almaty: ZIALY BASPASY, 2026.
[3] H. J. Wilson and P. R. Daugherty, "Collaborative intelligence: Humans and AI are joining forces," Harvard Business Review, vol. 96, no. 4, pp. 114–123, 2018.
[4] Bureau of National Statistics of the Republic of Kazakhstan, "Statistical indicators of commercial transport maintenance and retail trade in Astana," stat.gov.kz, 2026.

FIELD 6 · CAPSTONE LINK

Weekly Record 4 establishes the governance boundaries and human accountability hierarchy that directly feeds Capstone Section 4: 'Human–AI Operating Model'. The review points and automation thresholds determined here will be integrated into the 12-month transformation roadmap defended in the final presentation.

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WEEKLY INTEGRATED LEARNING RECORD 5 (REDUCED)

Student Name: Manat Amir
Group: DPA-2602
Course: Business Administration (BA 1204)
Academic Year: 2026–2027
Focal Org (Company B): AB Auto (Astana, Kazakhstan)
Comparator Org (Company A): Interkom-Auto (Russia / CIS)
Topic: Business Performance: Operations & Supply Chain (Ch. 10)
Attestation Period: 1st Attestation

FIELD 1 · CONCEPT (CART Lens: Concept)

Operations management transforms inputs (labor, raw materials, capital, information) into finished market outputs. Key operational concepts include:
• **Production Planning:** Designing the overall production approach, facility location, layout, and resource scheduling.
• **Process Types:** Mass production (standardized high volume), customization (unique per client), and mass customization (standard core with customized features).
• **Just-In-Time (JIT):** A lean operational philosophy minimizing holding costs by receiving inventory only as needed for fulfillment.
• **Critical Path Method (CPM):** The longest sequence of dependent activities determining the minimum time required to complete an entire operational cycle.
• **Total Quality Management (TQM) & Lean:** Continuous pursuit of zero defects and relentless elimination of non-value-added waste (muda).
• **Bottleneck:** The specific stage or resource with the lowest capacity that restricts the total throughput of the entire operational system.
• **KPI (Key Performance Indicator):** A quantifiable metric tracking operational velocity, defect rates, or customer fulfillment standards [1], [2].

Concept Check — "Why does automating a non-bottleneck station not increase total output?":
Under the Theory of Constraints (Goldratt) and queuing theory, the maximum output of an entire linear system is strictly governed by the capacity of its slowest operational node (the bottleneck). Automating an upstream or downstream non-bottleneck simply produces work-in-progress (WIP) that piles up directly in front of the unaddressed constraint or creates idle starvation downstream. In the Paper Airplane Factory simulation, speeding up Station 1 created inventory queues at Station 2 without yielding a single additional accepted plane. Output rises only when the bottleneck itself is expanded or eliminated [1].

FIELD 2 · APPLICATION (Process Map, KPI, Controls & Fallback)

Step 1: Chosen Process Context

• **Process Name:** Urgent B2B Commercial Fleet Spare Parts Fulfillment
• **Trigger:** WhatsApp parts inquiry or counter walk-in from an STO repair mechanic for

urgent GAZelle parts.
• **Result:** Accurate, verified part handed over at Koktal 15B counter; financial transaction or credit balance logged.
• **Why it Matters:** Commercial vehicle downtime costs fleet operators upwards of 25,000 KZT/day; rapid counter fulfillment preserves B2B mechanic loyalty against marketplaces [4].

Step 2: Current State Process Map (Today — Bottleneck Identified)

#	Actor (Swimlane)	Step (Verb + Object)	Time / Waiting	Bottleneck / Observed Problem
1	**Customer (Mechanic)**	Submits parts request via WhatsApp / counter	1–3 min	Vague descriptions; missing VIN / OEM numbers.
2	**Sales Manager**	Manually cross-references paper supplier catalogs	5–10 min	Slow analogue lookup across fragmented manufacturer lists.
3	**Sales & Storekeeper**	Searches Koktal bins & calls Abylai Khan branch	15–35 min	**• BOTTLENECK: Stock unsynchronized across 4 IPs; physical aisle search.**
4	**Storekeeper**	Retrieves part from bin and brings to counter	5–8 min	Disorganized shelving causes search friction and misplaced boxes.
5	**General Manager**	Checks paper credit notebook for debt balance	5–10 min	Manual ledger checks cause checkout delays and math disputes.
6	**Customer & Staff**	Completes handover and signs paper receipt	2–4 min	Risk of handing out wrong sub-variant without digital check.

Step 3: The Target KPI

• **KPI Name:** Order-to-Dispatch Lead Time (Urgent B2B Counter Fulfillment)
• **Definition:** Total elapsed minutes from inquiry receipt to physical customer handover at the counter.
• **Baseline Today:** 65 minutes (empirical morning peak observation average).
• **Future-State Target:** Under 15 minutes (with 90% of in-stock lines fulfilled in <10 minutes).

Step 4: Future State Process Map (Redesigned with Allocation Categories)

#	Actor (Swimlane)	Step (Verb + Object)	Allocation Category	Impact on KPI
1	**Customer & AI Bot**	Submits part photo; AI extracts VIN & OEM code	**AI Executes + Human Review**	Cuts intake lookup time from 3 min to 30 sec.
2	**Cloud Inventory**	Queries real-time stock across Koktal, Abylai Khan & vendors	**Automated**	ELIMINATES BOTTLENECK: Drops lookup from 35 min to 5 sec.

3	**Storekeeper**	Picks part via mobile handheld bin routing code	**Automated (routing)**	Removes aimless search; picking under 4 minutes.
4	**Senior Storekeeper**	Verifies OEM stamp & physical integrity at counter	**Human + AI**	CRITICAL REVIEW POINT: Pre-handover check in <2 min.
5	**General Manager**	Approves credit line via automated credit limit score	**Human Only / Human + AI**	Replaces paper notebook; authorization in <1 min.
6	**Customer & Staff**	Scans QR code for instant digital invoice & pickup	**Automated**	Final handover and 1C sync in under 60 seconds.

Step 5: The Required Operational Controls (Rubric Rows 5 & 7)

Control Element	WHO (Named Role)	WHAT / WHEN (Trigger)	HOW FAST (Time Limit)
Human Review Point	Senior Storekeeper (Supervisory)	Inspects OEM barcode, casting stamps, and gasket seal integrity against bill of materials before parts are boxed.	Under 2 minutes per order.
Escalation Path	Store Manager -> General Manager	Triggered if AI recognition confidence is <85%, price variance >±10%, or client disputes credit balance.	Resolved within 15 minutes.
Manual Fallback (Monday 09:00 Outage)	Operations Lead & Counter Staff	If internet/1C fails: switch to pre-printed order tickets, manual bin index, and paper emergency credit ledgers.	Immediate fallback; digital sync within 12h.